LOVELY PROFESSIONAL UNIVERSITY
School of Computer Science & Engineering | Department of AI & ML

INT395: SUPERVISED LEARNING - CAPSTONE PROJECT REPORT

CrowdPulse: Multi-Agent Crowd Density Estimation and Coordinated Alerting System

Problem Statement ID: #20 | **Target Dataset:** ShanghaiTech Part B
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1. Executive Summary

Crowd safety management at high-density public events and transport hubs requires early detection of potential crush conditions. Standard deep neural network approaches (e.g., CSRNet, MCNN) suffer from high compute latency and memory requirements, making them impractical for local edge deployment on embedded smart camera micro-controllers. **CrowdPulse** proposes a lightweight, interpretable, and modular architecture based on classical spatial feature engineering and Scikit-learn regression ensembles.

The system partitions surveillance frames into a 3x3 spatial grid (9 zones), extracts 31 domain-specific visual descriptors (HOG orientation bins, Canny edge density ratio, LBP uniform texture histograms, and pixel intensity statistics), and estimates localized head counts using a Random Forest regression pipeline. These continuous counts are categorized into operational risk tiers (Green: <10, Yellow: 10-24, Red: ≥25), evaluated with SHAP explainability, and monitored by decentralized autonomous Zone Agents capable of triggering coordinated bottleneck alerts and dynamic rerouting instructions before dangerous crushes form.

2. INT395 Course Outcome (CO) Alignment

Outcome	Syllabus Requirement	CrowdPulse Implementation
CO1	Problem Formulation	Formulated crowd monitoring as localized spatial regression across 9 zones.
CO2	Data Preprocessing & Features	Parsed ShanghaiTech .mat files, extracted 31 HOG/Canny/LBP descriptors.
CO3	Model Formulation	Implemented Ridge, SVR(RBF), HistGradientBoosting, and Random Forest.
CO4	Cross-Validation & Tuning	5-fold CV and hyperparameter optimization for variance reduction.

CO5	Performance Evaluation	Continuous metrics (MAE 6.54, R2 0.73) & 3-Tier Risk Confusion Matrix (94.4% acc).
CO6	Deployment & XAI	SHAP TreeExplainer beeswarm/waterfalls and Streamlit live telemetry dashboard.

3. Mathematical Methodology & Feature Descriptors

Each surveillance frame is sliced into 9 zones ($Z_{r, c}$) with row index $r \in \{0, 1, 2\}$ and column index $c \in \{0, 1, 2\}$. From each normalized 256x256 patch, 31 spatial visual features are extracted:

- **HOG Orientations (10 features):** 8 gradient orientation bins (0°-180°) capturing head/shoulder contour angles, plus mean and std gradient magnitude.
- **Canny Edge Density & Gradient (2 features):** Ratio of edge pixels to patch area ($R = \frac{1}{|P|} \sum \mathbb{I}_{\text{edge}}(p)$) and mean Sobel intensity on edge pixels.
- **Local Binary Patterns (12 features):** 10 uniform texture bins ($P=8, R=1$) distinguishing human hair/clothing from smooth pavement, plus Shannon entropy and energy.
- **Pixel Intensity Statistics (5 features):** First and second order moments: Mean, Standard Deviation, Variance, Skewness, Kurtosis.
- **Normalized Spatial Coordinates (2 features):** Vertical and horizontal position ($r/2, c/2$) encoding camera perspective scaling.

4. Experimental Results & Model Benchmarking

Models were evaluated using 5-fold cross-validation on the 3,600 training zone samples and verified against the 2,844 test zone samples:

Model Architecture	CV MAE	CV RMSE	CV R^2	Hold-Out Test MAE
Ridge Regression (Baseline)	13.62	28.30	0.434	13.62
SVR (RBF Kernel)	7.99	25.63	0.535	7.99
HistGradientBoosting	7.06	19.83	0.722	7.06
Random Forest (Champion)	**6.54**	**19.65**	**0.727**	**6.54**

5. Operational Safety Risk Classification

Discretizing continuous predictions into 3 operational risk tiers achieves **94.37% overall accuracy** and **0.763 macro F1-score** on hold-out test data:

Risk Tier	Count Range	Precision	Recall	F1-Score	Action Protocol
Green (Low)	< 10 heads	0.995	0.983	0.989	Normal pedestrian transit.
Yellow (Moderate)	10 - 24 heads	0.568	0.351	0.434	Pacing caution & monitoring.

Red (High)	≥ 25 heads	0.795	0.953	0.867	Bottleneck alert & rerouting.

6. Multi-Agent Coordination & Alert Broker

Each zone is governed by an autonomous **ZoneAgent** connected to an 8-neighbor spatial adjacency graph managed by the **AgentCoordinator**:

1. **Isolated Red Zone:** Generates a localized caution to restrict immediate intake.
2. **Adjacent Red Zones:** When two connected zones (e.g., Zone_0_0 and Zone_0_1) reach the Red Tier simultaneously, the coordinator escalates to a **CRITICAL COORDINATED BOTTLENECK ALERT** and automatically computes pedestrian relief paths diverting crowd traffic toward neighboring Green corridors.

7. Conclusion

CrowdPulse successfully proves that classical supervised regression ensembles combined with principled spatial feature engineering can deliver highly accurate, explainable, and real-time crowd density monitoring on resource-constrained edge hardware without the prohibitive overhead of monolithic deep neural networks.